(Fig. 3). The gap should be .002"-.004". If the gap setting is incorrect, reset the gap adjusting screw.

8. On governors without the starting aid screw, hold the gap adjusting screw and tighten the locknut.

9. Recheck the gap with the engine operating between 1100 and 1300 rpm. Readjust, if necessary.

10. Stop the engine and install the fuel rod between the differential lever and the control tube lever.

11. Use a new gasket and install the governor cover and lever assembly.

Position Injector Rack Control Levers

The position of the injector racks must be correctly set in relation to the governor. Their position determines the amount of fuel injected into each cylinder and ensures equal distribution of the load.

Properly positioned injector rack control levers with the engine at full load will result in the following:

1. Speed control lever at the maximum speed position.

2. Governor low-speed gap closed.

3. High-speed spring plunger on the seat in the governor control housing.

4. Injector racks in the full-fuel position.

> **NOTE:** When positioning the injector racks on an early engine equipped with a fuel modulator, be sure no interference is encountered from the fuel modulator. Loosen the fuel modulator lever ("U" bolt) and move the modulator lever along the injector control tube to avoid contact with the modulator cam or the adjacent cylinder head stud nut.

Adjust the No. 1 injector rack control lever first to establish a guide for adjusting the remaining injector rack control levers (Fig. 4).

1. Disconnect any linkage attached to the governor speed control lever.

2. Turn the idle speed adjusting screw until 1/2" of the threads (12-14 threads) project from the locknut when the nut is against the high-speed plunger (Fig. 3). This adjustment lowers the tension of the low-speed spring so it can be easily compressed. This permits closing the low-speed gap without bending the fuel rods or causing the *yield mechanism springs to yield or stretch.*

> **NOTE:** A false full fuel rack setting may result if the idle speed adjusting screw is not backed out as noted above.

Injector racks must be adjusted so that the effort to move the throttle from the *idle speed* position to the *maximum speed* position is uniform. A sudden increase in effort can result from:

a. Injector racks adjusted too tight, not allowing the speed control lever to reach the end of its travel.

b. Binding of the fuel rod.

c. Failure to back out the idle screw.

3. Back out the buffer screw approximately 5/8" if it has not already been done.

4. Loosen all of the inner and outer injector rack control lever adjusting screws or adjusting screws and locknuts. Be sure all the injector rack control levers are free on the injector control tube.

> **NOTE:** On engines equipped with a yield link type fuel rod, attach a small "C" clamp at the shoulder of the rod to prevent the yield spring from compressing while adjusting the injector rack control levers.

5. Move the speed control lever to the maximum speed position (Fig. 4). Hold the lever in that position with light finger pressure and proceed as follows:

Two Screw Assembly -- Turn the inner adjusting screw on the No. 1 injector rack control lever down until a slight movement of the control tube is observed or a step-up in effort is noted. This will place the No. 1 injector rack in the full-fuel position. Turn the outer adjusting screw down until it bottoms lightly on the injector control tube. Then, alternately tighten both the inner and outer adjusting screw.

One Screw and Locknut Assembly -- Tighten the adjusting screw of the No. 1 injector rack control lever until the injector rack clevis is observed to roll up or an increase in effort to turn the screwdriver is noted. Tighten the screw approximately 1/8 of a turn more and lock securely with the adjusting screw locknut. This will place the No. 1 injector rack in the full-fuel position.

> **NOTE:** Overtightening of the injector rack control lever adjusting screws during installation or adjustment can result in damage to the injector control tube. The recommended torque of the adjusting screws is 24-36 **lb-in** (3-4 Nm).

The above step should result in placing the governor linkage and control tube assembly in the same position

© 1984 General Motors Corp.